Arithmetic and complexity

Deciding accurate evaluability of real polynomials

Difficulty: extreme **Importance:** broadly interesting

Status: Solved

Rating rationale: Extreme because the existence of any accurate computation tree is a general decidability problem beyond checking a supplied algorithm; broad importance includes numerical stability, symbolic computation and automated algorithm design.

Independently reviewed resolution - 2026-09-11

Theorem 1.1 gives an always-halting decision procedure for the exact constant-free finite-tree model in the statement. In every signed ordering chart $x_{\pi(i)} = \sigma_i(y_1 + \cdots + y_i)$, the coefficientwise absolute majorant of $q(y) = p(x(y))$ must be bounded by $C|q(y)|$ on $y \geq 0$. This finite real-quantifier criterion is necessary and sufficient; Sections 2-4 prove the equivalence and construct an evaluator when it holds. Both independent reviews cover comparisons, branching, stored-value reuse and arbitrary independent rounding errors.

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A second independent proof review also passes. The described experimental programs were not retained; their run counts and compiler/solver claims are not reproduced or certified here. The full mathematical proof, rather than those logs, supports the resolution. The ratings and rationale above are historical assessments of the original open target.

Problem statement

Consider finite arithmetic computation trees with exact real inputs $x \in \mathbb{R}^n$. Arithmetic nodes use binary addition, subtraction, or multiplication, each returning $(a \circ b)(1+\delta_j)$. Negation and comparisons $(<, =, >)$ of available quantities are exact, and branching is allowed. There are no constant input nodes. Each operation may have a different, arbitrary error δ_j; equal operands need not produce equal errors. Previously computed values may be stored and reused. The tree describes control flow, not a formula that recomputes each occurrence of a subexpression.

For $p \in \mathbb{Z}[x_1, \ldots, x_n]$ with $p(0) = 0$, call such a tree P accurate if $P(x, 0) = p(x)$ and

$$\forall \eta \in (0,1)\ \exists u \in (0,1) \quad \forall x \in \mathbb{R}^n\ \forall \delta \in [-u, u]^{N(P)} : \quad |P(x,\delta) - p(x)| \leq \eta |p(x)|,$$

where $N(P)$ numbers all rounded nodes; errors on unused branches are ignored. The same tree must work for every η, and u may depend on p, P, η but not on x. The condition includes exact output at zeros of p.

Question

Is there a Turing algorithm which, from the finite integer coefficient list of any such p, always halts and decides whether an accurate tree exists? No bound on the decision algorithm's running time is prescribed.

This is a basic obstruction to automatically constructing accurate algorithms for determinants and other polynomial expressions associated with structured matrices. Checking a supplied fixed tree is already decidable; deciding whether any suitable tree exists is the question.

References and status

J. Demmel, I. Dumitriu, O. Holtz, and P. Koev, *Accurate and Efficient Expression Evaluation and Linear Algebra*, *Acta Numerica* 17 (2008), 87–145, Definitions 3.3–3.4, §3.3 and especially §3.3.7. The finite-tree, integer-coefficient formulation fixes the traditional real-arithmetic version of their decision question. Theorem 3.12 answers the corresponding whole-space complex question, not the real one. The original Demmel–Dumitriu–Holtz paper, *Toward accurate polynomial evaluation in rounded arithmetic*, v2, §§2.2–2.7, explicitly specifies uniformly bounded running time, exact comparisons, and separate rounding errors, clarifying the computation-tree conventions. Demmel's 21 October 2025 Simons seminar abstract still identifies the broader accurate-evaluation decision procedure as open. Searches on 2026-09-08 for accurate polynomial evaluability, real-arithmetic decidability, and follow-ups to the cited authors found no complete decision procedure or undecidability theorem for the displayed model. The recent talk does not separately specify every tree convention used here.

Status check — 2026-09-10

Rechecked Demmel et al., §3.3.7 and Demmel's October 2025 Simons abstract, and searched for later real-polynomial decision procedures. The broad real accurate-evaluation question remains explicitly open in the seminar abstract; no complete algorithm or undecidability theorem for the displayed finite-tree model was located. The complex-domain characterization and fixed-tree verification do not answer this existence question.